

Day 19 – ATS Documentation & Knowledge Transfer

Technical documentation, architecture, scoring logic, troubleshooting notes, and developer guidance

Objective	To make ATS maintainable, explainable, and extendable.
Primary Deliverable	ATS technical documentation and knowledge-transfer guide.
System Focus	ATS scoring, candidate shortlisting, decision logic, API behavior, and troubleshooting.

Prepared as the Day 19 knowledge-transfer artifact for the Zecpath AI hiring workflow.

1. Objective & Scope

The Day 19 objective is to make the ATS maintainable, explainable, and extendable. The documentation consolidates the implementation knowledge needed by a developer or reviewer to understand how ATS evaluation connects to recruitment decisions.

Detailed Tasks

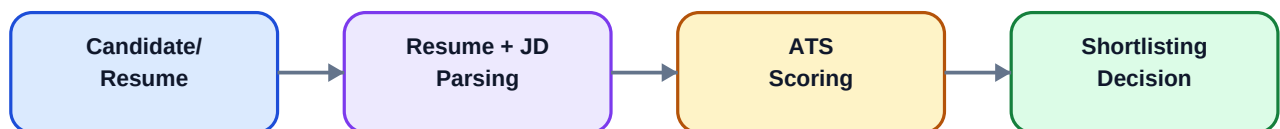
- Write technical documentation.
- Prepare architecture diagrams.
- Document scoring logic.
- Create troubleshooting notes.

Deliverables

Deliverable	Completed Knowledge Area
ATS technical documentation	System flow, components, inputs, outputs, and API behavior
Architecture diagrams	Resume/JD processing → ATS scoring → shortlisting decision flow
Developer guide	Setup guidance, endpoint behavior, scoring interpretation, and validation

2. ATS Architecture & System Flow

The documented workflow connects candidate data and job requirements to ATS evaluation and then to a recruitment decision. The shortlisting API exposes the resulting decision in a structured response.



Structured API response: candidate_id • ats_score • status • message

Architecture Notes

- Candidate and job information are transformed into structured evaluation inputs.
- The ATS scoring layer produces a numerical score used by the recruitment workflow.
- The shortlisting layer evaluates the score against configured decision criteria.
- The API returns a consistent response containing the candidate identifier, ATS score, status, and completion message.

3. ATS Scoring Logic Documentation

The scoring layer converts candidate-job alignment into a numerical ATS score. The resulting score is then consumed by the shortlisting workflow. Keeping the scoring and decision layers conceptually separate makes the system easier to maintain and extend.

ATS score	Decision interpretation	Documented test example
High score	Eligible for stronger consideration / shortlist path	Higher score should satisfy the configured threshold
Mid-range score	Recruiter review	60 → review
Low score	Reject	40 → rejected

Maintainability Guidance

- Keep score calculation rules configurable rather than hard-coding business decisions into unrelated components.
- Record the model/configuration version used when a score is produced.
- Use consistent candidate identifiers so scores and recruitment decisions remain traceable.
- Validate representative score scenarios after changing scoring or decision rules.

4. Shortlisting API & Validation

The documented shortlisting workflow was validated using candidate scenarios with different ATS scores. The API response structure was checked to ensure the decision status matched the tested score scenario.

Test Scenario A — ATS 60 → Review

```
{
  "candidate_id": "CAND-82624710",
  "ats_score": 60,
  "status": "review",
  "message": "Candidate shortlisting completed"
}
```

Test Scenario B — ATS 40 → Rejected

```
{
  "candidate_id": "CAND-82624710",
  "ats_score": 40,
  "status": "rejected",
  "message": "Candidate shortlisting completed"
}
```

API Response Contract

Field	Purpose
candidate_id	Unique identifier used to track the candidate.
ats_score	Numerical ATS score produced by the scoring engine.
status	Recruitment decision returned by the shortlisting logic.
message	Confirms completion of the candidate shortlisting operation.

5. Troubleshooting Notes

Problem area	Recommended check
Blank React page	Check browser Console for runtime errors, then verify the affected component/import and route.
Route not rendering	Verify the Day component import and its <Route path=...> entry in App.jsx.
PDF not opening	Confirm the PDF exists in the public folder and that the browser path begins with /.
Unexpected ATS decision	Verify the input score, configured decision criteria, and the latest scoring configuration.
API response mismatch	Compare candidate_id, ats_score, status, and message against the expected response contract.

6. Developer Guide

A developer maintaining the ATS workflow should treat scoring, decision rules, API contracts, and documentation as connected but separately testable layers.

- **Understand the flow:** candidate/job data → parsing/structuring → ATS scoring → decision logic → API response.
- **Change safely:** update scoring or decision configuration deliberately and rerun representative score scenarios.
- **Validate outputs:** confirm the returned `candidate_id`, `ats_score`, `status`, and `message`.
- **Keep documentation current:** update architecture and troubleshooting notes whenever the workflow or API contract changes.

Knowledge Transfer Checklist

- Developer can explain the ATS-to-shortlisting workflow.
- Developer can identify the key response fields.
- Developer can interpret review and rejection test scenarios.
- Developer can troubleshoot route, PDF, scoring, and API-response issues.
- Developer understands where scoring/decision changes should be documented and validated.

Day 19 Outcome: The ATS workflow is documented as a maintainable knowledge-transfer artifact, covering architecture, scoring logic, API behavior, troubleshooting, and developer guidance.